

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PHY 413 — Past Questions & Materials
by Renance

Course: PHY 413
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 400

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Paper 1 of 2 — PHY 413 Electromagnetic Fields and Waves

Level 400 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS

SECOND SEMESTER EXAMINATION: 2023/2024 Academic Session

. COURSE CODE: PHY 413 ' P

COURSE TITLE: Electromagnetic Fields and Waves.

Instructions: You are expected to answer Question | and any other 3. All the symbols retain their usual meaning. ; : ,

TIME ALLOWED: 3 hrs.

COURSE UNIT: 3.

e *

Question 1. Define the following terms as applied to electricity and magnetism.

(a) What is a field? (b) List two types of fields (c) State two Questions for each type of fields. (b) What is electromagnetic 'field (e) List two characteristics of electromagnetic' field (f) What is wave? (g) State three characteristics of waves (h) Name two types of waves (i) State two Questions for each type of wave (j) What is electromagnetic wave (k)

List seven types of electromagnetic wave (1) List five characteristics of electromagnetic wave.

Question 2. Consid

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in medium 2, if we use the symbols i, r and i, respectively to label quantities refer these waves, then show that the following quantities.

a) E,) _ (Z,cos@, -Z,cos6,. (b) Ey) a. 2Z, cos8,
E, mit Z,cos@, +Z, cos@, E,), Z,008@, +Z, cos,

t i /

Question 3. The appropriate equation describing the propagation of plane waves in a conducting medium for any component of the electric field E and magnetic induction B is : :

2 2

e Bae - nee v2 0

az? at ar?

(a) Define all the symbols which appear in the equation and their various units.

(b) By using a plane wave of the form; $w(z,t) = y, e^{i(kz - \omega t)}$ " ? ~ " show that

$\tan Q = \frac{f}{\omega} \left| \frac{+QP}{-@Q} \right|$

(c) If $Q=2$, calculate Q.

Consider two charge points P and P' in space specified by the position

Question 4. $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Now use any known vector

vectors $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$

technique to show that

$\nabla \cdot \mathbf{r} = 3$

0

: 2

: $\nabla \cdot \mathbf{r} = 3$

(b) $\nabla \cdot \mathbf{r} = 3$

7 pr located in

Question 5. Consider a system of N point charges, $q_1, q_2, \dots, q_N$ in

some volume of space V'. The position vector of the charge are $\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N$ in a

the possible angle between any of the charges is θ then use the multiple expansion

technique to show that the potential $V(r)$ due to the point charges is

N

$\nabla \cdot \mathbf{r} = 3$

$g(r) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \frac{q_i}{r_i}$

. i=

Question 4 (2) State Maxwell's equations in integral form

and explain the equation and their various units.

and define all the symbols which

appear in Maxwell's displacement current equations.

(c) Use the Maxwell's equations in integral form

in vacuum (Q6.a) to show that the electric field vector

$\mathbf{E}$ is given by

$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^N \frac{q_i \mathbf{r}_i}{r_i^3}$

$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$

at $\mathbf{r}$?

Define all the symbols!

(4) Show that $\nabla \cdot \mathbf{r} = 3$

Paper 2 of 2 — Electromagnetic fields and waves - PHY 413

Level 400 • First Semester • Handout

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DEPARTMENT OF PHYSICS

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Question 1. Define the following terms as applied to electricity and magnetism.

(a) What is a field? (b) List two types of fields (c) State two Questions for each type of fields. (d) What is electromagnetic field (e) List two characteristics of electromagnetic field (f) What is wave? (g) State three characteristics of waves (h) Name two types of waves (i) State two Questions for each type of wave (j) What is electromagnetic wave (k) List seven types of electromagnetic wave (l) List five characteristics of electromagnetic wave.

Question 2. Consider the propagation of three waves across a certain medium. The incidence wave in medium 1, a reflected wave also in medium 1, and a transmitted wave in medium 2, if we use the symbols i , r and t , respectively to label quantities referring to these waves, then show that the following quantities.

(a) $\left(\frac{E_r}{E_i}\right)_{\perp} = \left(\frac{Z_2 \cos \theta_i - Z_1 \cos \theta_t}{Z_2 \cos \theta_i + Z_1 \cos \theta_t}\right)$ (b) $\left(\frac{E_t}{E_i}\right)_{\perp} = \left(\frac{2Z_2 \cos \theta_i}{Z_2 \cos \theta_i + Z_1 \cos \theta_t}\right)$

Question 3. The appropriate equation describing the propagation of plane waves in a conducting medium for any component of the electric field E and magnetic induction B is

$$\frac{\partial^2 \psi}{\partial z^2} - \mu\sigma \frac{\partial \psi}{\partial t} - \mu\epsilon \frac{\partial^2 \psi}{\partial t^2} = 0$$

(a) Define all the symbols which appear in the equation and their various units.
 (b) By using a plane wave of the form; $\psi(z, t) = \psi_0 e^{i(kz - \omega t)}$ show that

$$\tan \Omega = \sqrt{1 + Q^2} - Q$$

(c) If $Q=2$, calculate Ω .

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Question 4. Consider two charge points P and P' in space specified by the position vectors $\vec{r} = xi + yj + zk$ and $\vec{r}' = x'i + y'j + z'k$. Now use any known integral technique to show that

$$(a) \quad \int_0^{\infty} \frac{a^2 r^2}{(r^2 + a^2)^{5/2}} dr = \frac{1}{3}$$

$$(b) \quad \nabla^2 \left(\frac{1}{\sqrt{r^2 + a^2}} \right) = -\frac{3a^2}{(r^2 + a^2)^{5/2}}$$

Question 5. Consider a system of N point charges, $q_1, q_2, q_3, \dots, q_i, \dots, q_N$ located in some volume of space V' . The position vector of the charge are $\vec{r}_1, \vec{r}_2, \vec{r}_3, \dots, \vec{r}_i, \dots, \vec{r}_N$. If the possible angle between any of the charges is θ_i then use the multiple expansion technique to show that the potential $\varphi(r)$ due to the point charges is

$$\varphi(r) = \frac{1}{4\pi\epsilon_0 r} \sum_{i=1}^N \left(q_i + \frac{1}{r} q_i r_i \cos\theta_i + \frac{1}{2r^2} q_i r_i^2 (3\cos^2\theta_i - 1) + \dots \right)$$

Question 6. (a) State Maxwell's equations in a vacuum and define all the symbols which appear in the equation and their various units.

(b) State Maxwell's displacement current equations.

(c) Use the Maxwell's equations in a vacuum (Q6.a) to show that the electric field vector $\vec{E}$ is given by

$$\nabla^2 \vec{E} = \mu\sigma \frac{\partial \vec{E}}{\partial t} + \mu\epsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

Define all the symbols which appear in the equation and their various units.

(d) Show that when no sources $\phi = 0$ are present the electric field vector $\vec{E}$ is

$$\vec{E} = -\frac{1}{c} \frac{\partial \vec{A}}{\partial t}$$